AG952 | Textual Analytics for Accounting and Finance

Workshop: BrewDog in the News: Reading the Signals

Student Handout

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A Note on this handout:

Keep this alongside the Colab notebook as you work. It gives you the background you need to make sense of what you are doing at each stage and includes spaces for you to record your choices and observations. Come back to the blank sections after the session to consolidate what you found. Your analyst's note at the end (CP6) draws directly on the thinking you do in those boxes.

PART ONE: THE BREWDOG CASE STUDY

BrewDog was founded in 2007 by James Watt and Martin Dickie out of a small, rented unit in Fraserburgh, Aberdeenshire. Their opening offer to the world was Punk IPA, a beer that was deliberately more bitter, more heavily hopped, and more aggressively marketed than anything sitting on a British supermarket shelf at the time. Within a year they had won a handful of competition medals and secured a listing with Tesco. Within two years they had attracted enough attention from the craft beer press to generate serious national newspaper coverage.

What made BrewDog unusual as a corporate story was not just the beer. In 2010, the company launched "Equity for Punks", an equity crowdfunding scheme offered directly to retail investors at a time when this kind of thing was still a novelty. Investors received shares, discounts on beer, and the opportunity to say they owned part of a craft brewery. The press loved it. Over successive rounds, more than 200,000 individuals invested, raising more than £75 million in aggregate. It was one of the largest crowdfunding exercises ever run by a private company in the UK.

The 1,011 newspaper articles in your corpus span roughly 15 years. The notebook groups them into five narrative eras:

(i) Growth (2010 – 2014)

BrewDog was growing fast and the press coverage reflected it. Most articles in this period are about the beer itself, the expansion of the bar network, and the Equity for Punks model. The company was widely positioned as an irreverent success story. Financial distress vocabulary is almost entirely absent from this period.

(ii) Scaling (2015 – 2018)

In June 2017, private equity firm TSG Consumer Partners acquired a minority stake, valuing BrewDog at approximately £1 billion. This brought a new type of press coverage: financial analysis, M&A commentary, and questions about whether the "punk" identity would survive institutional investment. International expansion accelerated, with a brewery opening in Columbus, Ohio. Coverage volume increases noticeably, and the vocabulary shifts towards corporate finance terms alongside the beer and bars content that was dominant before.

(iii) Controversy (2019 – 2021)

This is the most significant inflection point in the corpus. In June 2021, more than 60 former BrewDog employees published an open letter describing the company as "a culture of fear". Allegations included bullying, discrimination, and a working environment that had left people with lasting psychological damage. James Watt denied some of the specific claims but acknowledged problems with the company culture. Coverage volume spiked sharply. Critically for your analysis, the language in this period is often negative but not financially negative. Words like "fear", "toxic", and "bullying" carry clear negative sentiment in everyday use, but they do not appear in financial dictionaries constructed from SEC filings. This is one of the central tensions you will explore today.

(iv) Decline (2022 – 2023)

By late 2022, BrewDog's auditors had flagged going-concern risk in the annual report. This is the formal accounting signal that a company may not be able to continue operating, and it is rare enough to attract serious press attention. Revenue was falling, energy costs were rising, and the post-pandemic return to pubs and bars had not rescued margins as the company had hoped. This period brings genuinely financial distress vocabulary into the corpus for the first time: "going concern", "refinancing", "impairment", "debt covenant".

(v) Administration (2024 – 2025)

James Watt announced his departure as chief executive in March 2024. Coverage since then has been dominated by questions about succession, asset sales, and whether the company can avoid formal administration. At the time of writing, the company continues to trade, but the arc of the story is clearly unresolved.

PART TWO: THE RESEARCH QUESTIONS

You are working on three linked research questions throughout the session. Keep them in mind as you move through each checkpoint.

RQ1: How does the topic composition of media coverage change across BrewDog's growth, controversy, and crisis phases?

RQ2: Do different sentiment methods agree on the timing and direction of financial distress signals in the corpus?

RQ3: Are the predictions of transformer-based sentiment models interpretable and trustworthy enough for use in regulated financial analysis?

PART THREE: STEP-BY-STEP GUIDE

STEP 0: Getting started

Run the Step 0 cell first. It installs all dependencies and takes about 90 seconds. Then run the Configuration cell. Do not skip this -- the submission cell at the end needs it.

Checkpoint 0: Team registration

Select your team name from the dropdown and click Register. This records your team in the session and unlocks CP1.

Checkpoint 1: Corpus exploration

Click "Load corpus". The cell loads your 1,011 articles and produces four charts: articles per year (colour-coded by narrative era), articles by newspaper, article length distribution, and mean article length by era.

Look at the articles-per-year chart carefully. You will see four annotated events marking key moments in the story. Run the CP1 Feedback cell before moving on.

What does the shape of coverage volume tell you before you look at any text? Note which years see the sharpest spikes and try to match them to the events annotated on the chart.

The Sun and The Times between them account for nearly three-quarters of the corpus. What might this concentration mean for any sentiment scores you calculate? Think about the difference between tabloid and broadsheet coverage of the same corporate events.

Checkpoint 2: Pre-processing choices

This is the first decision point with real methodological consequences. You have four choices to make:

Normalisation method: None, Porter Stemmer, or Lemmatisation

Stop-word list: NLTK standard, Finance-adjusted, or None

Remove numbers: Yes or No

Lowercase: Yes or No

The recommended combination for this corpus is Lemmatisation with the Finance-adjusted stop-word list. The reason will become clear in a moment. When you click

"Apply pipeline", the cell shows you a word cloud, a vocabulary statistics summary, and a vocabulary consequence display showing which negation and modal words your stop-word choice removes or retains. Run the CP2 Feedback cell. Pay attention to the explanation of why Finance-adjusted is recommended for financial text.

Which normalisation method did your team choose, and why?

The vocabulary consequence display shows which negation words (not, never, without, cannot) your stop-word choice removes from the corpus. Why does this matter for a sentiment analysis that will follow later? What would happen to the sentence "The company did not benefit from the restructuring" under NLTK standard stop-word removal?

CHECKPOINT 2b: Document clustering

Before running LDA, you will run an unsupervised k-means clustering step. This assigns each of the 1,011 articles to exactly one cluster based on TF-IDF term similarity. Unlike LDA, which produces a probability distribution over topics per article, k-means gives a hard assignment: each article belongs to one cluster and one cluster only.

Set your value of k using the slider (the default is 6, which works well for this corpus), then click "Run clustering". The process has four steps and takes about 20 to 30 seconds. You will see:

(i) A table showing cluster ID, the five highest-weight TF-IDF terms at the centroid (the cluster's automatic label), article count, and share of the corpus.

(ii) A representative headline: the single article closest to each cluster centre by cosine similarity.

(iii) A t-SNE scatter plot showing all 1,011 articles projected into two dimensions, coloured by cluster. Tight, well-separated regions suggest the corpus organises cleanly by topic. Overlapping regions contain topically ambiguous articles.

(iv) A stacked bar chart showing the narrative era composition of each cluster. A cluster dominated by a single era likely reflects a time-specific story rather than a perennial theme.

List the cluster labels your model produced (the five-term centroid descriptions). Can you interpret what each cluster represents? Try to give each one a human-readable name.

Look at the era composition chart. Are any clusters concentrated in a single narrative era, or are they spread across the full timeline? What does this tell you about whether the corpus organises by topic or by time period?

CHECKPOINT 3: LDA topic modelling

Set your number of topics k using the slider, then click "Run LDA". The model runs on the full 1,011-article corpus. After fitting, it prints the top 10 words per topic. Read the word lists carefully before entering your topic labels. The text boxes at the bottom prompt you to give each topic a short name. This is a genuine interpretive task - there is no correct answer, and teams in the same class often label the same statistical topic quite differently.

After saving your labels, use the two additional buttons:

"Compare eras" fits LDA separately on each narrative era and produces a heatmap. The chart uses a log base-2 relative frequency scale: a value of +1 means a term appeared twice as often in that era as on average across the full corpus; -1 means half as often; 0 means exactly at the corpus average. Red cells are era-defining vocabulary. White cells are background noise common to all eras.

"STM-style analysis" shows how topic prevalence shifts over time, using a line chart and an OLS slope for each topic.

What value of k did your team choose, and what was the topic distinctiveness score (reported below the topic word lists)? Run the CP3 Feedback cell before moving on?

k = _______ Distinctiveness score = _______ %

Why did you choose this value of k?

Which topic showed the steepest rising or falling trend in the STM-style analysis? Does the direction of that trend make sense given what you know about the BrewDog story?

CHECKPOINT 4: Sentiment analysis

This is the core of the workshop. Choose your model from the dropdown:

(i) Dictionary only (VADER and Loughran-McDonald, instant)
(ii) FinBERT -- finance-tuned transformer
(iii) DistilBERT-SST2 -- generic transformer

If the session is time-pressured, dictionary only is a reasonable choice. If you have a GPU runtime, FinBERT is much faster and pedagogically more interesting. After inference completes, use the four buttons above the output area to explore different views of the results:

"Sentiment over time" plots all three methods on the same chart across the full 2010 to 2025 window.

"By newspaper" shows mean VADER scores by publication. Note that these are not era-controlled, so read them with care.

"By theme" breaks down sentiment by the LDA topic assigned to each article.

"Show disagreements" (when a transformer has been run) identifies the articles where VADER and the transformer disagree most strongly. These are the analytically interesting cases.

Run the CP4 Feedback cell before moving on.

Look at the "Sentiment over time" chart. At what point does each method first show a sustained negative reading? Does the Loughran-McDonald dictionary, VADER, and the transformer agree on the timing of the turn, or does one method pick it up earlier than the others?

Examine the disagreements table (articles where VADER and the transformer disagree strongly). Pick one example from each direction (VADER positive, transformer negative; and VADER negative, transformer positive) and write the headline and your explanation for why the two methods diverged on this article.

CHECKPOINT 5: Integrated gradients

This section uses a technique called integrated gradients to inspect what the transformer model is attending to inside each prediction. For a given input sentence, it attributes a score to each word that reflects how much that word shifted the model's output from a neutral baseline. Positive attribution (green) pushes the prediction towards positive; negative attribution (red) pushes it towards negative.

Work through the four buttons in order:

Button A: two straightforward sentences (clear positive and clear negative) that serve as a sanity check.

Button B: two tricky sentences where the financial and the everyday meaning of words diverge. Sentence 3 mentions "administration" and "risk" in a context where both risk and administration are being resolved. Sentence 4 is about workplace culture, not financial performance.

Button C: Strobe mode (runs all five sentences automatically).

Button D: runs integrated gradients on the top three articles from your CP4 disagreements table. This is the most analytically interesting button as it connects the corpus-level disagreements you identified in CP4 to the token-level explanations.

Select your answer to the interpretability reflection question in the notebook, then run the CP5 Feedback cell.

For sentence 3 ("The company's auditors noted that the risk of administration had been materially reduced following last month's refinancing agreement"), which tokens received the highest attribution scores? Did the model attend to the words that you, as a reader, would consider most important for interpreting this sentence?

Sentence 4 is about workplace culture and contains no financial distress vocabulary. How did the model classify it, and does the attribution pattern explain the prediction in a way you would trust?

CHECKPOINT 6: Analyst's note

Write a maximum of 150 words addressed to an investment committee. Your note must cover three things:

1. Whether the text analysis methods, taken together, provide evidence of early warning signals in the BrewDog corpus.
2. The most significant limitation that would affect the reliability of those signals.
3. What additional data or analysis you would need before including this in a formal investment memorandum.

Click "Submit note" when you are ready. Your note will appear in the class debrief.

PART FOUR: POST-SESSION REFLECTION

These questions are for your own consolidation after the workshop. There are no right answers. They are designed to help you think carefully about what the session means for how you use these tools in practice.

Q1. The Loughran-McDonald dictionary was constructed from SEC filings written by lawyers and accountants. VADER was trained partly on news and social media. FinBERT was fine-tuned on Bloomberg and Reuters financial news. Given those different origins, what kind of text is each method best suited to, and where might each one fail.

Q2. The culture open letter (June 2021) is unambiguously negative in tone. A general-purpose sentiment model should detect that. But from the perspective of an equity investor in BrewDog, is a culture scandal the same kind of negative signal as a going-concern audit note? How should a financial text analysis system handle the difference between reputational risk and solvency risk?

Q3. Integrated gradients gave you a token-level explanation of each prediction. For regulated financial analysis under the EU AI Act, explanation is not optional...it is a requirement. Based on what you saw in CP5, would you consider the explanations produced by FinBERT sufficient to satisfy that standard? What would "sufficient" even look like in this context?